

BASAVARAJESWARI GROUP OF INSTITUTIONS

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

Autonomous Institute under VTU, Belagavi | Approved by AICTE, New Delhi Recognized by
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NAAC Accredited Institution*
(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to
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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Course Project Report

“Social Media Addiction Analyzer”

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Visvesvaraya Technological University
Belagavi, Karnataka

2025-2026



Department of Artificial Intelligence & Machine Learning

Academic year 2025-26

Rubrics for Course Project

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Course Code:22AI463

Batch No. : 1

Project Title
Social Media Addiction Analyzer

USN	Name	Problem Identification & Objective Formulation (5)	Preprocessing Dataset Collection & (5)	Application of Data Analytics Techniques (7)	Visualization & Interpretation of Results (4)	Project Presentation Report & Viva (4)	Total Marks (25)
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**DEPARTMENT OF
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING**

CERTIFICATE

This is to certify that the Course Project entitled “ **Social Media Addiction Analyzer** ” has been successfully completed for the course Introduction to Data Analytics by {**MOHAMMED AFAAZ, MOHAMMED IRFAN, MOHAMMED TALHA, MOHAMMED YOUSUF, SAI KIRAN S**} bearing {**3BR24AI092, 3BR24AI094, 3BR24AI098, 3BR24AI100, 3BR24AI140**}, students of IV semester B.E. for the partial fulfillment of the requirements for the Bachelor of Engineering in Artificial Intelligence & Machine Learning during the academic year 2025-26.

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ABSTRACT

The Social Media Addiction Analyzer is a data-driven system developed to analyze the impact of excessive social media usage on students' productivity, mental health, sleep quality, and academic performance. Using a Kaggle dataset of 705 students with 13 demographic and behavioral attributes, the data was preprocessed through cleaning, duplicate removal, and feature selection to ensure analytical reliability. Interactive Power BI dashboards provide insights into addiction distribution, platform-wise usage, mental health correlations, demographic patterns, and behavioral trends, revealing that a majority of students fall into the high addiction category and that higher social media usage is associated with poorer mental health and academic outcomes. The dashboard spans four report pages covering KPI summaries, platform risk analysis, conflict escalation patterns, geographic usage distribution, and a platform-by-gender insights matrix. The findings demonstrate how business intelligence and visual analytics can support digital well-being awareness and informed decision-making in educational environments.

PROBLEM STATEMENT

To design and develop a dashboard that analyzes the impact of social media usage among students on academic performance, mental health, sleep quality, and productivity using Power BI.

1. INTRODUCTION

The rapid proliferation of social media platforms has fundamentally transformed how young people communicate, learn, and spend their time. Platforms such as Instagram, TikTok, YouTube, Facebook, and WhatsApp have become deeply embedded in the daily routines of students and adolescents across the world. While these platforms offer genuine benefits in terms of connectivity, self-expression, and information access, their excessive and unregulated use has emerged as a serious public health concern.

Social media addiction is characterized by compulsive and uncontrollable engagement with social networking sites to a degree that interferes with daily functioning, academic responsibilities, and mental well-being. The addictive design features of these platforms — including infinite scrolling, push notifications, likes, and algorithmic content curation — are engineered to maximize engagement, often exploiting psychological vulnerabilities in young users. This has led to a situation where a significant proportion of students find themselves unable to regulate their usage even when they are aware of its harmful effects.

The consequences of this addiction are wide-ranging. Studies have consistently shown a negative correlation between social media addiction and mental health outcomes, including elevated levels of anxiety, depression, loneliness, and diminished self-esteem. Sleep deprivation is another critical consequence, as late-night scrolling and screen exposure disrupt natural sleep cycles, which in turn further deteriorates mental health and academic focus. Academically, students addicted to social media are more likely to experience poor concentration, procrastination, lower grades, and reduced motivation.

In India specifically, the situation is particularly concerning. Indians spend an average of 2.4 hours per day on social media, with young people aged 18 to 24 being the most active demographic. Facebook and Instagram alone account for 97.2 million and 69 million users from this age bracket in India respectively. Furthermore, approximately 85 percent of Indian children have reported experiencing cyberbullying, a rate more than twice the international average.

Despite the scale of this problem, there exists a critical lack of personalized, data-driven tools that can help individual students, educators, and counselors understand the severity of social media habits at both the population and individual level. This project addresses that gap by developing the Social Media Addiction Analyzer — a data analytics system anchored by an interactive Power BI dashboard and supported by a machine learning-powered web application. The dashboard provides aggregate visual analytics across the full 705-student dataset, enabling population-level trend exploration, while the web application translates those insights into individual addiction score predictions, productivity forecasts, and personalized recommendations.

2. OBJECTIVES

1. Identify the key behavioral and psychological factors contributing to social media addiction among students
2. Collect and preprocess the dataset by resolving null values, duplicates, and inconsistencies
3. Analyze patterns and correlations in the data to generate meaningful insights
4. Visualize the findings through a comprehensive multi-page interactive Power BI dashboard

3. DATASET COLLECTION

The dataset used in this project is titled Students Social Media Addiction and was sourced from Kaggle. It contains records from 705 students across multiple countries and academic levels, providing a diverse and representative sample for analyzing social media usage behavior and its associated outcomes.

Table 3.1. Dataset Variables

SL No.	Variable	Description
1	Student_ID	Unique identifier for each student record
2	Age	Age of the student (numeric)
3	Gender	Gender of the student (Male / Female)
4	Academic_Level	Current level of study (High School / Undergraduate / Graduate)
5	Country	Country of the student
6	Avg_Daily_Usage_Hours	Average number of hours spent on social media per day
7	Most_Used_Platform	Primary social media platform used (e.g., Instagram, TikTok, YouTube)
8	Affects_Academic_Performance	Whether social media use affects academic performance (Yes / No)
9	Sleep_Hours_Per_Night	Average number of hours slept per night
10	Mental_Health_Score	Self-reported mental health score (1–10 scale)
11	Relationship_Status	Relationship status (Single / In Relationship / Complicated)
12	Conflicts_Over_Social_Media	Number of conflicts caused by social media usage (0–5 scale)
13	Addicted_Score	Self-reported addiction score (1–9 scale) — primary regression target

A derived target variable, Productivity Label (Low / Medium / High), was engineered from a weighted combination of sleep hours, mental health score, daily usage, and conflict count, then binned into three equal percentile tiers for the productivity classification task.

4. SOFTWARE REQUIRED

- Microsoft Power BI Desktop — Primary analytics and dashboard visualization tool
- Microsoft PowerPoint — Creating presentation slides
- Microsoft Word — Creating project report

5. IMPLEMENTATION

(i) Data Preprocessing

The dataset was first inspected for data quality issues across all thirteen columns. One row containing null values was identified and removed, ensuring no imputation artifacts introduced bias into the analysis or training data. The dataset was also checked for duplicate Student IDs and row-level duplicates; none were found, confirming the integrity of the source data.

Categorical columns including Gender, Academic Level, Most Used Platform, Affects Academic Performance, and Relationship Status were standardized for consistent labeling. Within the Power BI pipeline, these columns were cleaned and formatted using Power Query Editor to ensure correct grouping, filtering, and cross-visual interaction. Within the Python machine learning pipeline, the same columns were transformed using dedicated LabelEncoder instances, and the encoder objects were serialized for consistent application at inference time.

Ten features were selected for model training: Age, Gender, Academic Level, Average Daily Usage Hours, Most Used Platform, Affects Academic Performance, Sleep Hours Per Night, Mental Health Score, Relationship Status, and Conflicts Over Social Media.

- **Data Validation:** The dataset was validated to ensure that all numerical values fell within their expected ranges and that categorical fields contained standardized values without inconsistencies, improving the reliability of subsequent analysis.
- **Power Query Transformations:** Power Query Editor was utilized to remove blank records, verify duplicate entries, format column data types, and create calculated columns before loading the processed data into the Power BI data model.
- **DAX-Based Analytics:** Custom DAX measures such as Average Screen Time, Average Addiction Score, High Risk Percentage, and Average Mental Health Score were created to provide dynamic KPI calculations that automatically update based on slicer selections and user interactions.
- **Interactive Filtering:** Slicers for Gender, Academic Level, Most Used Platform, and Relationship Status were implemented, enabling users to perform demographic-specific analysis and explore patterns across different student groups.

(ii) Formulae Used

1. **Average Daily Usage Hours:** $\text{Average Daily Usage} = \Sigma \text{Daily_Usage_Hours}_i / n$

Power BI DAX

Avg Screen Time = AVERAGE('Students Social Media Addiction'[Avg_Daily_Usage_Hours])

2. **Average Addiction Score:** $\text{Average Addiction Score} = \Sigma \text{Addicted_Score}_i / n$

Power BI DAX

Avg Addiction Score = AVERAGE('Students Social Media Addiction'[Addicted_Score])

3. **High Risk Percentage:** $\text{High Risk \%} = \text{No. of Students (Addiction Score} \geq 7) / (\text{Total Students}) \times 100$

Power BI DAX

High Risk % =

DIVIDE(

CALCULATE(

COUNTROWS('Students Social Media Addiction'),

'Students Social Media Addiction'[Addicted_Score] >= 7

),

COUNTROWS('Students Social Media Addiction')

) * 100

4. **Average Mental Health Score:** $\text{Avg Mental Health Score} = \Sigma \text{Mental_Health_Score}_i / n$

Power BI DAX

Avg Mental Health Score = AVERAGE('Students Social Media Addiction'[Mental_Health_Score])

5. **Category Frequency Calculation:** $\text{Category Count} = \Sigma I(\text{Student belongs to Category})$

where $I()$ is an indicator function

6. **Platform-wise Average Addiction Score:** $\text{Avg Score}_{\text{Platform}} = \Sigma (\text{Addicted_Score}) / (\text{No. of Students using that Platform})$

(iii) Dataset-Level Insights Generated

The following aggregate statistics were computed and stored for use in the dashboard and web application:

Table 5.1. Metric vs Value

SI No.	Metric	Value
1	Average daily usage	4.19 hours
2	Average addiction score	6.44 out of 9
3	Students reporting academic impact	64.3%
4	Low addiction tier (score 1–3)	43 students
5	Moderate addiction tier (score 4–6) High	234 students
6	addiction tier (score 7–9)	428 students (60.7%)
7	Platform with highest avg addiction score	WhatsApp / Snapchat(~7.4–7.5)
8	Pearson correlation: sleep vs addiction score	–0.12
9	Pearson correlation: daily usage vs addiction score	+0.15

6. DASHBOARD VISUALIZATIONS

The Power BI dashboard was built on a 705-student CSV dataset using Microsoft Power BI Desktop. It provides four interactive report pages, each designed to answer specific analytical questions about student social media behavior, addiction severity, and its downstream effects on mental health and academic life.

Overview & Platform Behavior

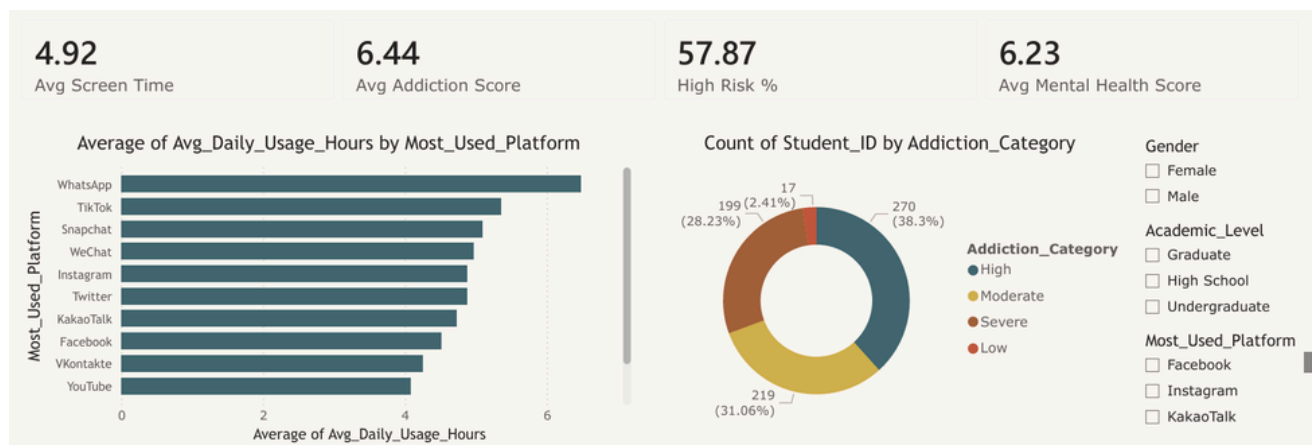


Figure 6.1 — Overview & Platform Behavior

Four KPI summary cards that establish the population-level baseline. A horizontal bar chart breaks down average daily usage by platform, where WhatsApp leads in time spent followed by WhatsApp, TikTok and Snapchat, while a doughnut chart visualizes addiction category distribution with the High tier accounting for the largest segment. A supporting data table provides a platform-level breakdown of student count, addiction scores, usage hours, and average mental health scores. A bar chart plots average addiction score against reported conflict frequency, revealing a near-linear escalation that establishes interpersonal conflict as a strong behavioral signal of addiction. A treemap visualizes total usage hours by country, with India and the USA occupying the largest segments.

Mental Health & Platform Risk

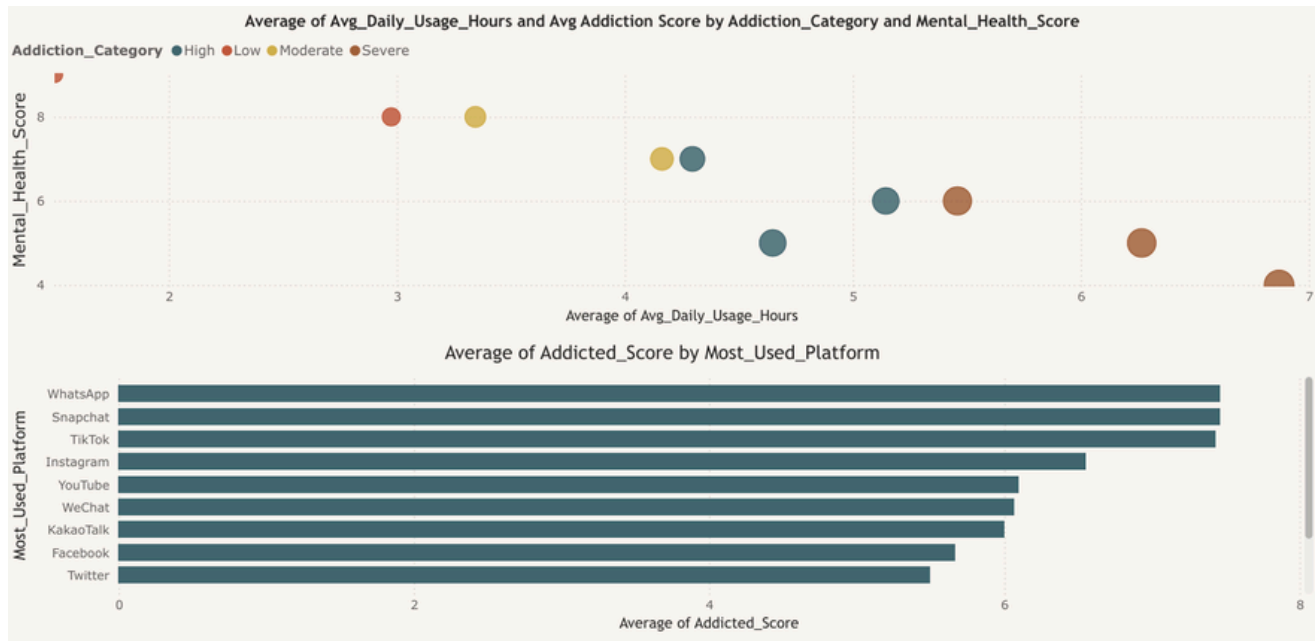


Figure 6.2 — Mental Health & Platform Risk

A scatter plot chart shows mental health score against average daily usage colored by addiction category, revealing that High and Severe students cluster at lower mental health scores even at moderate usage levels. A horizontal bar chart ranks platforms by average addiction score, with Snapchat and WhatsApp consistently at the higher end.

Conflict Escalation & Geographic Spread

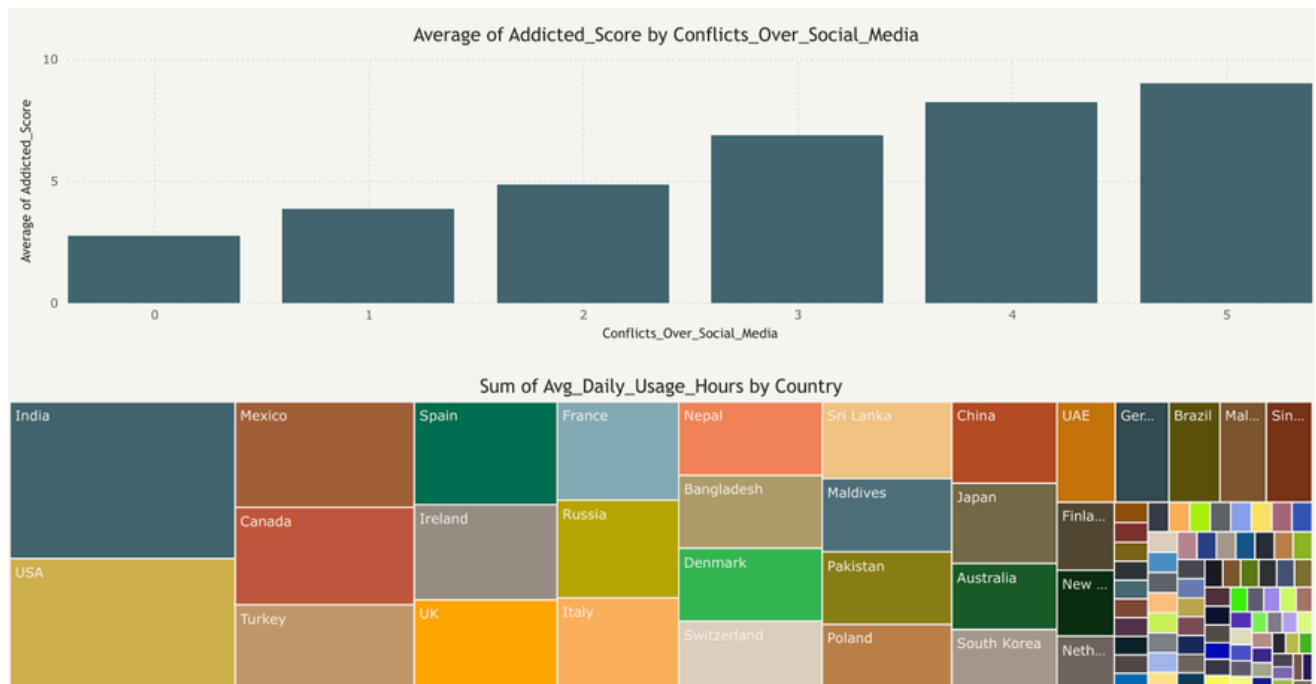


Figure 6.3 — Conflict Escalation & Geographic Spread

A bar chart plots average addiction score against reported conflict frequency, revealing a near-linear escalation that establishes interpersonal conflict as a strong behavioral signal of addiction. A treemap visualizes total usage hours by country, with India and the USA occupying the largest segments.

Scoring, Distribution & Platform-Gender Matrix

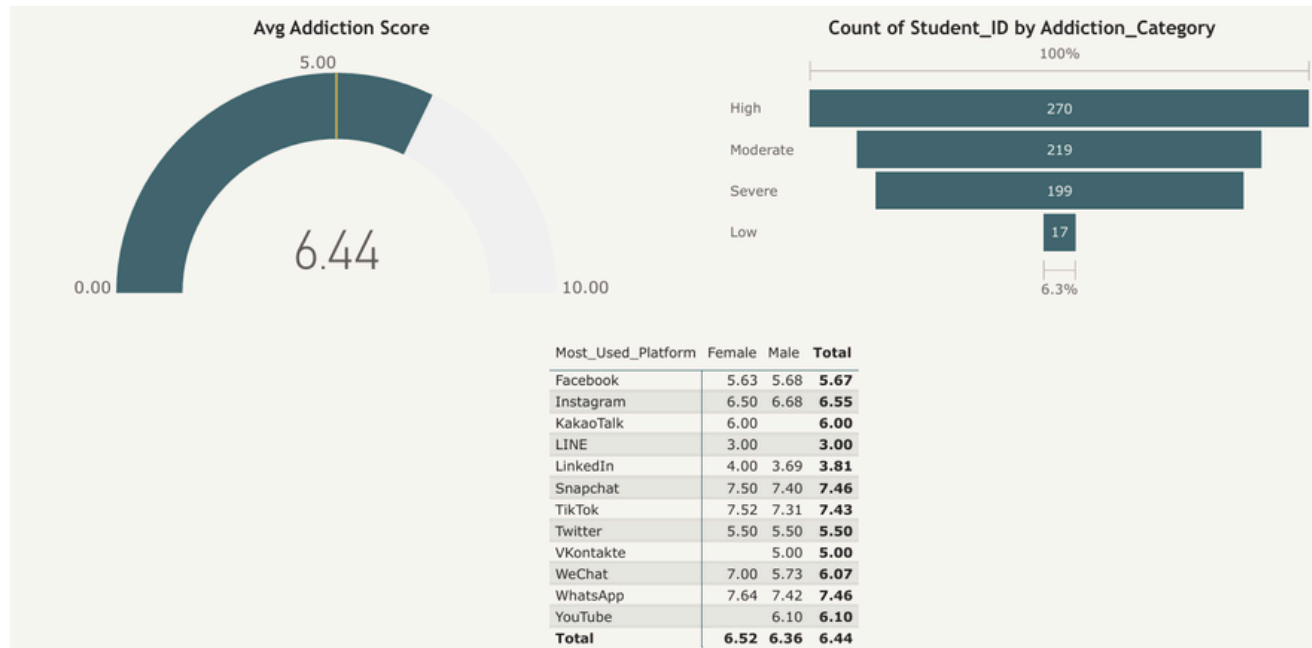


Figure 6.4 — Scoring, Distribution & Platform-Gender Matrix

The fourth page consolidates key findings into a focused view, featuring a gauge visual for the overall addiction score and a platform-by-gender matrix table where WhatsApp, TikTok, and Snapchat consistently show the highest scores across both genders.

7.RESULTS

The dashboard provides a consolidated view of social media addiction patterns across 705 students through KPI cards, platform comparisons, addiction category distribution, mental health correlations, conflict escalation trends, and a country-level treemap, with interactive filters for gender, academic level, and platform.

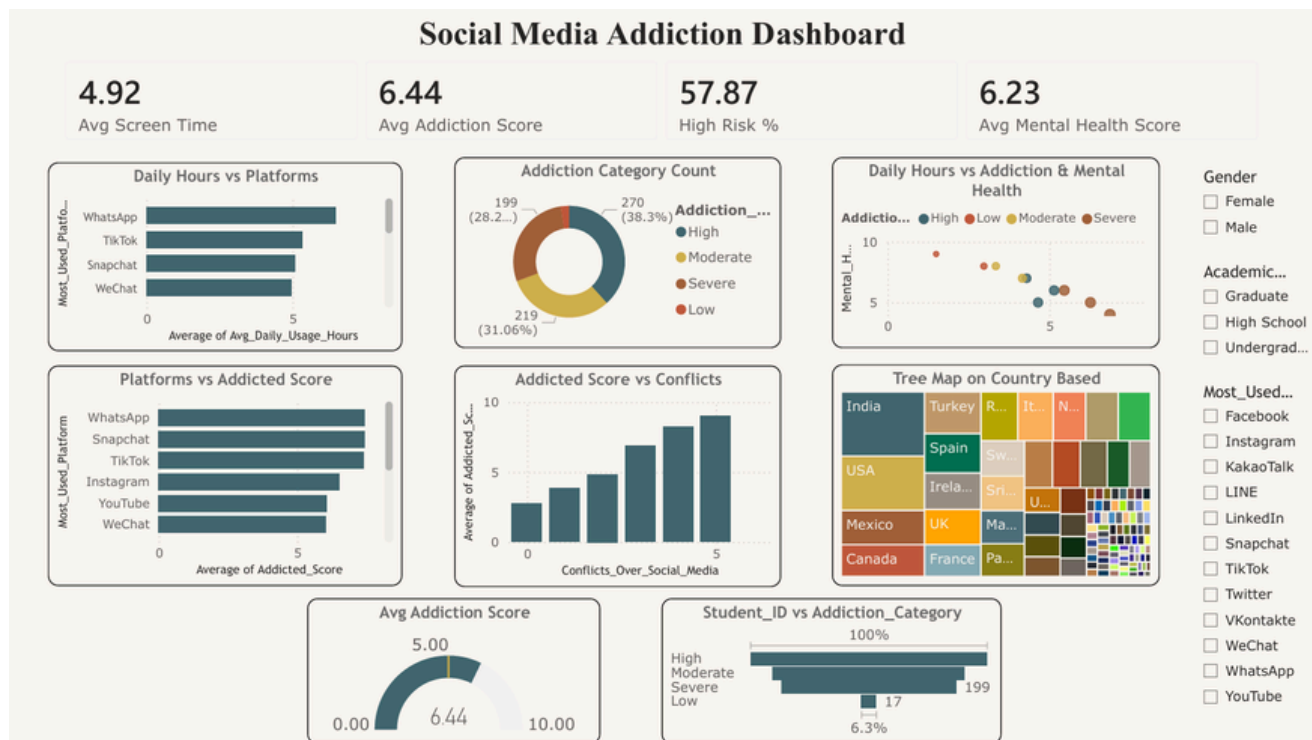


Figure 7.1. – Overall Power BI Dashboard

This Power BI dashboard visualizing social media addiction patterns across 705 students through KPI cards, platform comparisons, category distributions, and interactive filters.

Key Insights

- The High addiction category accounts for 38.3% of all students, making social media addiction a widespread behavioral pattern rather than an isolated concern
- WhatsApp, Snapchat, and TikTok record the highest average addiction scores, identifying them as the highest-risk platforms
- TikTok users show the lowest average mental health score among all platforms, highlighting a strong link between platform usage and psychological wellbeing
- Addiction score escalates consistently with increasing conflict frequency, confirming interpersonal conflict as a strong behavioral signal of addiction
- India and USA dominate the country-level treemap, representing the highest aggregate daily usage hours in the dataset

8. CONCLUSION

This project presents a comprehensive Social Media Addiction Analyzer built around a Power BI dashboard that delivers rich, interactive visual analytics across a 705-student dataset. By transforming raw behavioral and demographic data into structured, explorable visuals, the dashboard provides a clear and evidence-backed lens through which social media addiction patterns among students can be understood and communicated.

The Power BI dashboard stands as the core analytical contribution of this project. It translates raw student data into actionable visual insights — revealing that a majority of students fall in the high addiction tier, that TikTok and Snapchat carry the highest addiction risk among platforms, that a significant share of students experience academic disruption, and that higher addiction scores consistently correlate with reduced mental health scores and increased interpersonal conflict. These findings validate both the quality of the dataset and the effectiveness of the analytical approach.

The dashboard's multi-layered design — spanning KPI cards, platform comparisons, geographic treemaps, and category distributions — demonstrates how business intelligence tools can go beyond simple reporting to surface meaningful behavioral patterns at scale. The interactive filters further allow findings to be explored across gender, academic level, and platform dimensions, making the insights adaptable to a wide range of research and institutional contexts.

This project demonstrates the practical value of data analytics and business intelligence in understanding and communicating the behavioral patterns of social media addiction among students. Future work could extend the dataset with longitudinal tracking, incorporate additional variables such as study hours and physical activity, and deploy the dashboard as a cloud-hosted institutional tool accessible to counselors, educators, and academic advisors.

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